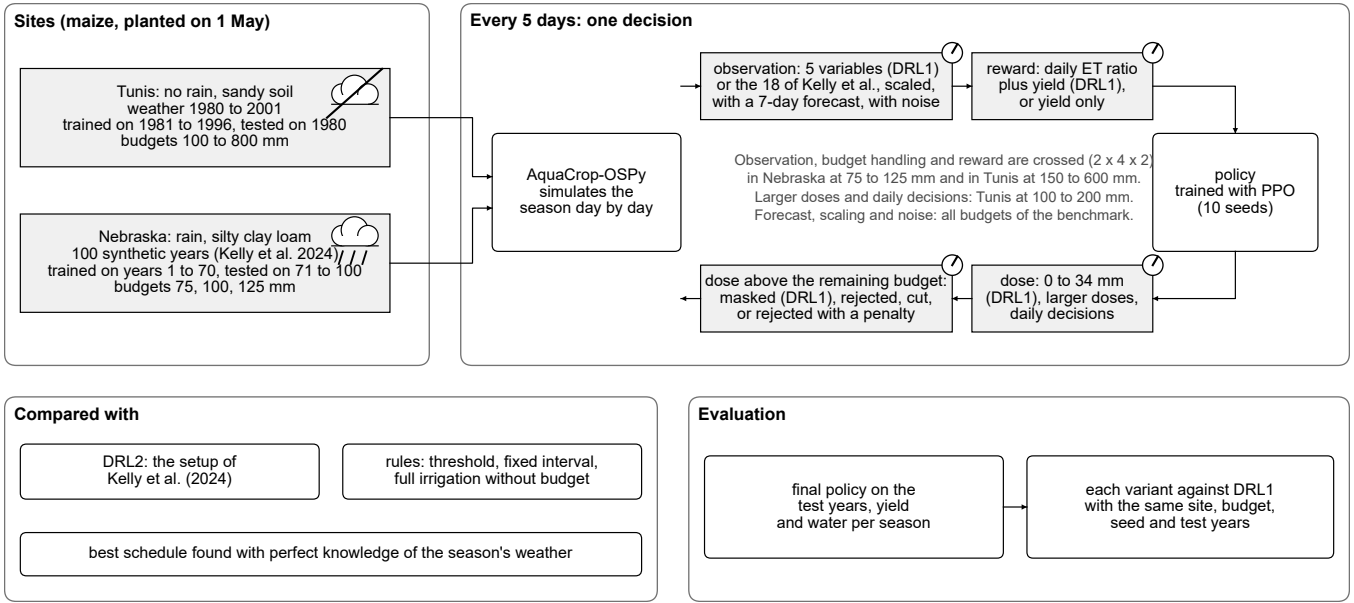
Sites (maize, planted on 1 May)
Tunis: no rain, sandy soil
weather 1980 to 2001
trained on 1981 to 1996, tested on 1980
budgets 100 to 800 mm
Nebraska: rain, silty clay loam
100 synthetic years (Kelly et al. 2024)
trained on years 1 to 70, tested on 71 to 100
budgets 75, 100, 125 mm
AquaCrop-OSPy
simulates
the season day by day
Every 5 days: one decision
observation: 5 variables (DRL1)
or the 18 of Kelly et al., scaled,
with a 7-day forecast, with noise
reward: daily ET ratio
plus yield (DRL1),
or yield only
policy
trained with PPO
(10 seeds)
Observation, budget handling and reward are crossed (2 x 4 x 2)
in Nebraska at 75 to 125 mm and in Tunis at 150 to 600 mm.
Larger doses and daily decisions: Tunis at 100 to 200 mm.
Forecast, scaling and noise: all budgets of the benchmark.
dose: 0 to 34 mm
(DRL1), larger doses,
daily decisions
dose above the remaining budget:
masked (DRL1), rejected, cut,
or rejected with a penalty
Compared with
DRL2: the setup of
Kelly et al. (2024)
rules: threshold, fixed interval,
full irrigation without budget
best schedule found with perfect knowledge of the season's weather
Evaluation
final policy on the
test years, yield
and water per season
each variant against DRL1
with the same site, budget,
seed and test years